Complements to 10: Making Ten & Carry Trade

Addition Strategies to 10 • Standard: CCSS.MATH.1.OA.C.6

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1. MATHEMATICAL CONCEPT & STACK APPROACH: Determine complements to 10 using Place Value Digit Blocks, Gold [+1] Carry Tokens, and RPN subtraction.

Our base-10 system regroups every collection of 10 ones into 1 ten. Finding a number's complement to 10 (10 - 7 = 3) places the target base (10) on Shelf Y and the current ones digit (7) on Workbench X.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Digit Blocks & Gold [+1] Carry Tokens)

Hands-On Setup: Place the [7] Digit Block into the Ones socket of the Place Value Tray and hold a Gold [+1] Carry Token beside the Tens socket.

Student Prediction (Before Calculator): How many more ones must be added to [7] to reach 10 and trigger the Gold [+1] Carry Token? (7 + 3 = 10).

3. CALCULATOR EXECUTION (SC-6 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [1] [0] [ENTER] [7] [-]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	Enter 1
[0]	10	Target ten (10) in X
[ENTER]	10	10 lifted to Y
[7]	7	Current Ones digit (7) in X
[-]	3	10 - 7 = 3: Adding 3 completes 10 and triggers the [+1] Carry!

4. MATHEMATICAL CONTEXT & REAL-WORLD MOTIVATION (Making a Ten First (8 + 5 = 10 + 3 = 13) & The 10-for-1 Carry Trade)

Why is 10 + 3 so much easier for our brains than 8 + 5? Because base-10 place value groups everything into full tens! When a column reaches 10 ones on the Place Value Tray, students trade all 10 ones for a single Gold [+1] Carry Token in the Tens column and leave the remaining ones in the Ones socket.

Core Investigation (Student Warm-Up A):

The Ones socket holds [6]. How many more ones are needed to complete a full 10 and trade for 1 Ten?

Bridging Through 10: The Make-Ten & Carry Strategy

Addition Within 20 • Standard: CCSS.MATH.1.OA.C.6

1. FOLLOW-UP EXTENSION CONCEPT, BLOCKS & PREDICTION

Concept & Stack Approach: Bridging across 10 decomposes the second addend so its first part completes a full ten (8 + 5 = (8 + 2) + 3 = 10 + 3). Completing the ten first turns any teen addition into an easy 10 + n fact.

Blocks & Prediction: Place [8] in Row 1 and [5] in Row 0 of the Ones column on the Place Value Tray. Prediction: Take 2 from the 5 to bring 8 up to 10. Place a Gold [+1] Carry Token in the Tens socket and [3] in the Ones socket: 10 + 3 = 13.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[8] [ENTER] [2] [+] [3] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[8]	**8**	8 in Ones socket
[ENTER]	**8**	8 lifted to Y
[2]	**2**	2 borrowed from 5 to make 10
[+]	**10**	8 + 2 = 10 (Gold [+1] Carry Token placed in Tens column!)
[3]	**3**	Remaining 3 ones
[+]	**13**	10 + 3 = 13 total ([1] Ten, [3] Ones)!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Counting on by ones across 10 (e.g., 8 + 5 as 9, 10, 11, 12, 13) instead of decomposing 5 into 2 + 3 to complete 10 first.

Hands-On Block Remediation: Have the student bridge [8] to 10 on the Place Value Tray, place a Gold [+1] Carry Token in the Tens socket with [3] in the Ones socket, and key [8] [ENTER] [2] [+] [3] [+].

Misconception 2: Reading teen numbers as two separate single digits (seeing 14 as 1 and 4 rather than 1 Ten plus 4 Ones).

Hands-On Block Remediation: Place [1] in the Tens socket and [4] in the Ones socket of the Place Value Tray and key [10] [ENTER] [4] [+] so students see 14 formed from 10 in Register Y and 4 in Register X.

Extension Challenge (Student Warm-Up B):

To add 9 + 6 using the Make-Ten strategy, split 6 into 1 and 5 so you make 10 first (9 + 1 + 5). What is the sum?

Expected Answer: 15 (9 + 1 = 10, then 10 + 5 = 15).

Complements to 10 & Make-10 Regrouping with Carry Tokens

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: COMPLEMENTS TO 10 & GOLD [+1] CARRY RULES

- The Ones column holds single Digit Blocks [0] through [9]. The moment a column reaches 10, it regroups!
- 10-for-1 Carry Trade: When the Ones column reaches 10, place a Gold [+1] Carry Token into the Tens column and reset the Ones socket to [0].
- Complements to 10: To find how much more is needed to trigger a Gold [+1] carry, enter 10 [ENTER], then the digit, then [-].

Key Sequence: `1` `0` `ENTER` `7` `-`

Investigation 1: You have 4 ones on your tray. How many more ones do you need to make a full 10 (4 + ? = 10)?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+, −, or ×):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 2: Solve 7 + 5 by bridging through 10: first add 7 + 3 to make 10, then add the remaining 2.

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+, −, or ×):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 3: The Ones socket holds [9]. How many more ones are needed to complete 10 and regroup into 1 Ten?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+, −, or ×):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 4: Solve 8 + 6 by splitting 6 into 2 and 4 so you make 10 first (8 + 2 + 4).

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+, −, or ×):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Unit 03: Complements to 10 & Make-10 Regrouping with Carry Tokens — Answer Key

Complete solutions and SC-6 verification keystrokes for Grade 1

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): The Ones socket holds [6]. How many more ones are needed to complete a full 10 and trade for 1 Ten?

Core Solution: 4 ones (10 - 6 = 4).

Core Check Keys: `1` `0` `ENTER` `6` `-`

Extension Challenge (Warm-Up B): To add 9 + 6 using the Make-Ten strategy, split 6 into 1 and 5 so you make 10 first (9 + 1 + 5). What is the sum?

Extension Solution: 15 (9 + 1 = 10, then 10 + 5 = 15).

Ext. Check Keys: `9` `ENTER` `1` `+` `5` `+`

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: You have 4 ones on your tray. How many more ones do you need to make a full 10 (4 + ? = 10)?

Expected Answer: 6

SC-6 Verification Keys: `1` `0` `ENTER` `4` `-`

Investigation 2: Solve 7 + 5 by bridging through 10: first add 7 + 3 to make 10, then add the remaining 2.

Expected Answer: 12

SC-6 Verification Keys: `7` `ENTER` `3` `+` `2` `+`

Investigation 3: The Ones socket holds [9]. How many more ones are needed to complete 10 and regroup into 1 Ten?

Expected Answer: 1

SC-6 Verification Keys: `1` `0` `ENTER` `9` `-`

Investigation 4: Solve 8 + 6 by splitting 6 into 2 and 4 so you make 10 first (8 + 2 + 4).

Expected Answer: 14

SC-6 Verification Keys: `8` `ENTER` `2` `+` `4` `+`

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.